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CASE STUDY:2

TOPIC: URBAN AIR QUALITY IN INDIAN CITIES – EXPLORATORY DATA ANALYSIS AND HYPOTHESIS TESTING

P24CS904

DATA SCIENCE

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CASE STUDY 2

URBAN AIR QUALITY IN INDIAN CITIES – EXPLORATORY DATA ANALYSIS AND HYPOTHESIS TESTING

Domain: Environment
SDGs: SDG 11 – Sustainable Cities and Communities; SDG 13 – Climate Action
Unit: II
Course Outcomes: CO2, CO6
Study Period: 2015–2020

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1. ABSTRACT

Air pollution is one of the major environmental challenges affecting urban areas in India. Rapid urbanisation, increasing vehicle usage, industrial activities, construction and population growth contribute to the deterioration of air quality. Exposure to harmful air pollutants such as Particulate Matter (PM2.5) and Nitrogen Dioxide (NO_2) can cause respiratory diseases, cardiovascular problems and other health-related complications.

This case study focuses on performing Exploratory Data Analysis (EDA) and statistical hypothesis testing on urban air quality data collected from Indian cities between 2015 and 2020. The study primarily investigates the distribution of PM2.5 and NO_2 concentrations across different cities and seasons. It also examines changes in air pollution during the COVID-19 lockdown period in 2020 compared with the corresponding period in 2019.

Python-based analytical tools, including Pandas, SciPy, Statsmodels and Seaborn, are used to perform data exploration, descriptive statistical analysis, confidence interval estimation, hypothesis testing and correlation analysis.

The study applies independent or paired statistical testing depending on the structure of the observations. It also investigates relationships between different pollutants and explains why statistical correlation does not necessarily indicate causation.

The findings are intended to provide insights into urban pollution patterns, seasonal variations and the relationship between human activities and environmental quality. This research supports sustainable urban development and climate action through evidence-based environmental analysis.

Keywords: Air Pollution, PM2.5, NO_2, Exploratory Data Analysis, Hypothesis Testing, Confidence Interval, COVID-19 Lockdown, Correlation Analysis, Indian Cities.

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2. INTRODUCTION

Air quality is an important indicator of environmental health and sustainable urban development. In India, many cities experience significant air pollution due to rapid industrialisation, transportation activities, construction and increasing energy consumption.

Air pollutants are harmful substances released into the atmosphere that negatively affect human health and the environment. Among the major pollutants, PM2.5 and NO_2 are particularly important because of their impact on respiratory health and atmospheric conditions.

Particulate Matter 2.5 (PM2.5) refers to airborne particles with a diameter of 2.5 micrometres or smaller. These fine particles can penetrate deep into the lungs and may enter the bloodstream. They are commonly generated through vehicle emissions, industrial activities, fuel combustion and other processes.

Nitrogen Dioxide (NO_2) is a gaseous air pollutant mainly associated with vehicle exhaust, power generation and industrial combustion. High concentrations of NO_2 can contribute to respiratory irritation and the formation of other atmospheric pollutants.

2.1 Importance of Air Quality Analysis

Air quality analysis helps researchers understand the environmental conditions of different cities. It provides information about pollution levels, seasonal variations and changes in pollutant concentrations over time.

The major reasons for conducting this study are:

1. To understand the distribution of PM2.5 and NO_2 across Indian cities.
2. To identify seasonal variations in air pollution.
3. To compare air pollution levels between 2019 and 2020.
4. To investigate the statistical significance of changes during the COVID-19 lockdown.
5. To identify relationships between different air pollutants.
6. To support environmental planning and pollution-control strategies.

2.2 COVID-19 Lockdown and Air Quality

The nationwide COVID-19 lockdown introduced in March 2020 significantly restricted transportation, industrial operations and several commercial activities.

These restrictions created an opportunity to investigate how changes in human activities were associated with air pollution levels.

Comparing corresponding dates in 2019 and 2020 helps researchers examine the differences in pollutant concentrations while reducing some seasonal variation.

However, meteorological conditions, rainfall, wind speed and other environmental factors must also be considered.

2.3 Sustainable Development Goals

SDG 11 – Sustainable Cities and Communities: This study supports the development of healthier cities through better understanding of urban air pollution.

SDG 13 – Climate Action: Environmental data analysis helps identify pollution patterns and supports evidence-based environmental management.

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3. PROBLEM STATEMENT

Urban air pollution has become a serious environmental issue in Indian cities. The concentration of harmful pollutants such as PM2.5 and NO_2 varies across cities and seasons due to differences in population density, transportation, industrial development and meteorological conditions.

Although air quality monitoring systems collect large amounts of environmental data, understanding these observations requires systematic statistical analysis.

The main problem addressed in this study is the lack of a unified exploratory and statistical comparison of PM2.5 and NO_2 distributions across Indian cities and seasons during 2015–2020.

The study also investigates whether the pollutant concentrations observed during the COVID-19 lockdown period in 2020 were statistically different from those recorded during the corresponding period in 2019.

Therefore, this project uses Exploratory Data Analysis, confidence interval estimation, hypothesis testing and correlation analysis to identify meaningful air quality patterns and provide evidence-based environmental insights.

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4. OBJECTIVES

The primary objective is to analyse air pollution patterns in Indian cities using statistical and exploratory data analysis techniques.

The specific objectives are:

1. To collect and examine historical air quality data from Indian cities covering 2015–2020.
2. To analyse the distribution of PM2.5 and NO_2 concentrations.
3. To compare pollutant concentrations across different cities.
4. To investigate seasonal variations in air pollution.
5. To calculate confidence intervals for pollutant concentration estimates.
6. To compare air pollution levels during the 2020 COVID-19 lockdown with the corresponding period in 2019.
7. To perform t-tests or Mann–Whitney tests to evaluate statistical differences.
8. To examine correlations between PM2.5, NO_2 and other available pollutants.
9. To understand why correlation does not establish a cause-and-effect relationship.
10. To provide environmental insights that support sustainable urban development.

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5. DATASET DESCRIPTION

The study uses historical urban air pollution measurements collected from monitoring stations across multiple Indian cities.

The selected dataset covers the period from 1 January 2015 to 31 December 2020 and contains daily and hourly measurements of several atmospheric pollutants.

The measurements originate from data provided by the Central Pollution Control Board (CPCB) and are made available through a compiled public dataset.

5.1 Dataset Source

Dataset Name: Urban Air Pollution Measurements, India, 2015–2020.

Original Dataset: Air Quality Data in India (2015–2020).

Source: Central Pollution Control Board (CPCB) measurements, compiled by the dataset contributor.

Dataset Link: https://www.kaggle.com/datasets/rohanrao/air-quality-data-in-india

Dataset Information: https://portal.transport-data.org/@tumi/urban-air-pollution-india-2015-2020

5.2 Cities Included

The dataset contains observations from several Indian cities, including:

- Delhi
- Mumbai
- Chennai

- Bengaluru
- Kolkata
- Hyderabad
- Ahmedabad
- Jaipur
- Lucknow
- Patna
- Bhopal
- Chandigarh
- Coimbatore
- Visakhapatnam

The availability of individual pollutants and observations may differ between cities and monitoring stations.

5.3 Important Dataset Attributes

Attribute	Description
Date	Date of measurement
City	Name of the Indian city
PM2.5	Fine particulate matter concentration
PM10	Coarse and fine particulate matter concentration
NO	Nitric oxide concentration
NO2	Nitrogen dioxide concentration
NOx	Nitrogen oxides concentration
SO2	Sulphur dioxide concentration
CO	Carbon monoxide concentration
O3	Ozone concentration
AQI	Air Quality Index

The primary variables selected for this study are PM2.5, NO_2, City and Date.

5.4 Study Period

The complete observation period is divided into the following analytical stages:

Period	Purpose
2015–2018	Historical pollution patterns
2019	Pre-lockdown comparison year
2020 (January–24 March)	Pre-lockdown period
2020 (25 March–31 May)	COVID-19 lockdown period
2020 (June–December)	Post-lockdown and reopening period

For the main lockdown comparison, 25 March to 31 May 2020 is compared with the exact corresponding calendar dates in 2019.

5.5 Dataset Suitability

The dataset is suitable because it provides historical pollution measurements, city-level information and multiple pollutant variables.

It supports temporal analysis, seasonal comparisons, descriptive statistics, confidence interval estimation and hypothesis testing.

Data limitation: Monitoring station coverage, reporting frequency and missing values can differ between cities and years. The analysis must account for these differences before drawing conclusions.

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6. TOOLS AND TECHNOLOGIES

Python is used as the primary programming language because it provides powerful libraries for data analysis, statistical computation and visualization.

6.1 Python

Python is an open-source programming language widely used in scientific computing and data science. It provides an easy-to-understand programming environment for processing environmental datasets.

6.2 Pandas

Pandas is used for data loading, data cleaning, grouping, filtering and statistical summarisation.

Major operations include:

- Reading CSV files.
- Handling missing observations.
- Filtering cities and dates.
- Calculating mean and median values.
- Grouping data by city and season.

6.3 SciPy

SciPy provides statistical functions for conducting hypothesis tests.

It is used for:

- Independent t-tests.
- Paired t-tests.

- Mann–Whitney U tests.
- Confidence interval calculations.
- Statistical significance testing.

6.4 Statsmodels

Statsmodels is used for statistical modelling and inference.

It supports:

- Confidence interval estimation.
- Statistical summaries.
- Regression analysis.
- Correlation significance testing.

6.5 Seaborn

Seaborn is used to create statistical visualisations.

It supports:

- Box plots.
- Distribution plots.
- Heatmaps.
- Scatter plots.
- Seasonal comparison graphs.

6.6 Technology Summary

Tool	**Purpose**
Python	Main programming language
Pandas	Data manipulation
SciPy	Hypothesis testing
Statsmodels	Statistical inference
Seaborn	Data visualization
Matplotlib	Graph plotting
Jupyter Notebook	Interactive analysis

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7. EXPLORATORY DATA ANALYSIS (EDA)

Exploratory Data Analysis is the process of examining a dataset to understand its structure, distribution, patterns and relationships before conducting advanced statistical analysis.

In this project, EDA is performed to understand the variations in PM2.5 and NO_2 concentrations across Indian cities.

7.1 Data Loading

The dataset is loaded into Python using the Pandas library.

```
import pandas as pd
import numpy as np

df = pd.read_csv("city_day.csv")

print(df.head())
print(df.shape)
print(df.info())
```

The initial examination identifies the number of observations, available columns and data types.

7.2 Data Cleaning

Data cleaning is performed to improve the quality and reliability of the dataset.

The following operations are carried out:

1. Identify missing values.
2. Remove duplicate records.
3. Convert date columns into datetime format.
4. Convert pollutant columns into numeric values.
5. Identify invalid or negative pollutant measurements.
6. Examine extreme observations.
7. Verify city names and measurement units.

Example:

```
df["Date"] = pd.to_datetime(df["Date"])

df["PM2.5"] = pd.to_numeric(
    df["PM2.5"], errors="coerce"
)

df["NO2"] = pd.to_numeric(
    df["NO2"], errors="coerce"
)

print(df.isnull().sum())
```

Missing values are not automatically replaced with zero because zero would represent an actual measurement rather than an unavailable observation.

7.3 Descriptive Statistical Analysis

Descriptive statistics provide a summary of the pollutant concentrations.

The following measures are calculated:

- Mean
- Median
- Standard deviation
- Minimum
- Maximum
- Quartiles

```
print(df[["PM2.5", "NO2"]].describe())
```

The mean indicates the average pollutant concentration, while the median represents the central observation.

The standard deviation measures how widely pollutant concentrations vary around the mean.

7.4 City-Wise Pollution Analysis

City-wise grouping is used to compare the pollution levels of different Indian cities.

```
city_analysis = df.groupby("City")[
    ["PM2.5", "NO2"]
].mean()

print(city_analysis)
```

This analysis helps identify cities with relatively higher or lower average concentrations.

A bar chart can be used to visualise city-wise pollution levels.

7.5 Seasonal Analysis

Air pollution changes according to weather conditions and seasonal activities.

The study divides the year into four broad Indian seasonal categories:

Season	**Months**
Winter	December–February
Summer / Pre-monsoon	March–May
Monsoon	June–September
Post-monsoon	October–November

The season variable is created using the date column.

```
def get_season(month):
    if month in [12, 1, 2]:
        return "Winter"
    elif month in [3, 4, 5]:
        return "Summer"
    elif month in [6, 7, 8, 9]:
        return "Monsoon"
    else:
        return "Post-Monsoon"

df["Season"] = df["Date"].dt.month.map(get_season)

season_analysis = df.groupby("Season")[
    ["PM2.5", "NO2"]
].mean()

print(season_analysis)
```

Seasonal comparisons help identify how pollutant concentrations change under different environmental conditions.

7.6 Data Visualization

The following visualisations are used:

1. Histogram – to understand pollutant distributions.
2. Box plot – to identify spread and extreme observations.
3. Bar chart – to compare cities.
4. Line chart – to examine changes over time.
5. Heatmap – to identify relationships between pollutants.
6. Scatter plot – to investigate pollutant associations.

These visualisations help communicate the major characteristics of the dataset.

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8. CONFIDENCE INTERVAL ESTIMATION

A confidence interval is a statistical range used to estimate an unknown population parameter based on sample observations.

In this project, confidence intervals are calculated to estimate the average PM2.5 and NO_2 concentrations.

A 95% confidence interval is selected for the analysis.

The interval provides a range of plausible values for the population mean under the assumptions of the statistical method.

8.1 Confidence Interval Formula

For a sample mean, the confidence interval can be expressed as:

95% CI = Sample Mean ± t-critical × Standard Error

Where:

- Sample Mean represents the average pollutant concentration.
- t-critical represents the critical value from the t-distribution.
- Standard Error measures the uncertainty of the sample mean.

8.2 Python Implementation

```
from scipy import stats
import numpy as np

values = df["PM2.5"].dropna()

mean = values.mean()
se = stats.sem(values)

ci = stats.t.interval(
    0.95,
    df=len(values)-1,
    loc=mean,
    scale=se
)

print("Mean:", mean)
print("95% Confidence Interval:", ci)
```

The same procedure can be repeated for NO_2.

8.3 Interpretation

A narrower confidence interval indicates greater precision in estimating the population mean, under the assumptions of the method.

A wider confidence interval indicates greater uncertainty.

For city-wise comparisons, confidence intervals should be calculated separately for each city rather than treating all city observations as one homogeneous population.

For highly skewed or serially correlated daily observations, bootstrap methods that respect the sampling structure may be more appropriate.

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9. HYPOTHESIS TESTING – 2019 VS 2020 LOCKDOWN

Hypothesis testing is used to determine whether an observed difference between two groups is statistically significant.

The primary objective is to examine whether PM2.5 and NO_2 concentrations during the 2020 lockdown period differed from their corresponding levels in 2019.

9.1 Study Period

2019 comparison period: 25 March–31 May 2019.

2020 lockdown period: 25 March–31 May 2020.

Using matching calendar dates helps reduce the effect of seasonal differences.

9.2 Null Hypothesis (H_0)

There is no statistically significant difference in the mean PM2.5 and NO_2 concentrations between the corresponding lockdown-period dates in 2019 and 2020.

9.3 Alternative Hypothesis (H_1)

There is a statistically significant difference in the mean PM2.5 and NO_2 concentrations between the corresponding periods in 2019 and 2020.

9.4 Significance Level

The significance level is set to:

$\alpha = 0.05$

Decision rule:

- If p-value < 0.05, reject the null hypothesis.
- If p-value ≥ 0.05, fail to reject the null hypothesis.

9.5 Paired t-Test

A paired t-test is suitable when the observations are meaningfully matched, such as the same city and corresponding calendar date across 2019 and 2020.

The test evaluates whether the average paired difference is significantly different from zero.

```
from scipy.stats import ttest_rel

t_stat, p_value = ttest_rel(
    data_2019,
    data_2020,
    nan_policy="omit"
)

print("T-statistic:", t_stat)
print("P-value:", p_value)
```

The two arrays must contain correctly aligned pairs without unmatched or incorrectly ordered observations.

9.6 Independent t-Test

An independent t-test is used when the observations come from two independent groups.

Welch's t-test is generally preferred when the two groups may have unequal variances.

```
from scipy.stats import ttest_ind

t_stat, p_value = ttest_ind(
    data_2019,
    data_2020,
    equal_var=False,
    nan_policy="omit"
)

print("T-statistic:", t_stat)
print("P-value:", p_value)
```

9.7 Mann–Whitney U Test

The Mann–Whitney U test is a non-parametric alternative for comparing two independent samples.

It is useful when the observations do not satisfy the assumptions required for a conventional independent t-test.

```
from scipy.stats import mannwhitneyu

u_stat, p_value = mannwhitneyu(
    data_2019,
    data_2020,
    alternative="two-sided"
)

print("U-statistic:", u_stat)
print("P-value:", p_value)
```

The Mann–Whitney test evaluates differences in distributions and should not automatically be interpreted as a test of medians unless additional distributional assumptions hold.

9.8 Statistical Decision

The resulting p-value is compared with the selected significance level of 0.05.

A statistically significant result indicates evidence of a difference between the selected groups.

However, statistical significance alone does not establish that lockdown restrictions caused the observed change.

The effect size and percentage change should also be reported.

9.9 Recommended Analytical Approach

For this study, the preferred procedure is to calculate city-level average pollutant concentrations for the matching dates and compare paired city-level observations between 2019 and 2020.

This reduces the risk of treating thousands of daily observations as independent when they are actually temporally related.

If several cities and pollutants are tested, multiple-comparison correction should also be considered.

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10. CORRELATION ANALYSIS

Correlation analysis is used to identify the statistical relationship between two or more environmental variables.

In this study, the relationship between PM2.5, NO_2 and other available pollutants is investigated.

10.1 Pearson Correlation

Pearson correlation measures the strength and direction of a linear relationship between two numerical variables.

Its coefficient ranges from -1 to +1.

- +1 indicates a perfect positive linear relationship.
- -1 indicates a perfect negative linear relationship.
- 0 indicates no linear correlation.

10.2 Spearman Correlation

Spearman correlation measures the strength and direction of a monotonic relationship using ranked observations.

It is useful when the data are skewed or when the relationship is not necessarily linear.

10.3 Python Implementation

```
import seaborn as sns
import matplotlib.pyplot as plt

pollutants = df[
    ["PM2.5", "PM10", "NO2", "SO2", "CO", "O3"]
]

correlation = pollutants.corr(
    method="pearson"
)

sns.heatmap(
    correlation,
    annot=True,
    cmap="coolwarm",
    fmt=".2f"
)

plt.title("Pollutant Correlation Matrix")
plt.show()
```

10.4 Interpretation of Correlation

A positive correlation between PM2.5 and NO_2 may indicate that both pollutants tend to increase together in certain observations.

A negative correlation indicates that higher values of one variable tend to occur with lower values of another.

However, a correlation does not establish that one pollutant directly causes a change in another.

10.5 Why Correlation Does Not Mean Causation

Correlation identifies an association between variables, whereas causation establishes that a change in one variable directly produces a change in another.

For example, PM2.5 and NO_2 may show a positive correlation because both pollutants are influenced by vehicle emissions.

However, their relationship may also be influenced by temperature, wind speed, rainfall, industrial activity and atmospheric chemistry.

Therefore, the observed relationship alone cannot prove that NO_2 causes an increase in PM2.5.

Additional research methods, environmental measurements and causal analysis are required to establish cause-and-effect relationships.

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11. RESULTS AND DISCUSSION

The results section presents the findings obtained from exploratory data analysis and statistical hypothesis testing.

11.1 Expected EDA Observations

The analysis is expected to identify differences in PM2.5 and NO_2 distributions across cities.

Cities with greater traffic density and industrial activity may show different pollutant profiles compared with cities having lower emission sources.

Seasonal analysis may reveal higher particulate concentrations during certain winter and post-monsoon periods, although the actual pattern can differ by city.

Monsoon rainfall and atmospheric dispersion may influence measured pollutant concentrations.

These patterns must be confirmed using the selected dataset.

11.2 Expected Lockdown Comparison

Previous published research provides evidence that air pollutant concentrations decreased in many Indian cities during the COVID-19 lockdown.

For example, Khan et al. (2021) reported average reductions of approximately 36% in PM2.5 and 58% in NO_2 during the lockdown across selected monitoring stations and comparison periods.

Another study examining five Indian cities reported varying PM2.5 reductions during lockdown, demonstrating that the magnitude of change differed between cities.

These published findings provide background evidence for the present analysis. They are not the statistical results of this project's own dataset.

11.3 Hypothesis Testing Results Table

The following table should be completed after executing the statistical analysis.

Pollutant	Test	Test Statistic	P-value	Decision
PM2.5	Paired t-test / alternative	To be calculated	To be calculated	To be determined
NO_2	Paired t-test / alternative	To be calculated	To be calculated	To be determined

11.4 Confidence Interval Results

Pollutant	Mean	Lower 95% CI	Upper 95% CI
PM2.5	Calculated value	Calculated value	Calculated value
NO_2	Calculated value	Calculated value	Calculated value

11.5 Correlation Results

The pollutant correlation matrix is used to identify positive, negative and weak relationships among the measured pollutants.

The analysis helps understand whether certain pollutants tend to vary together.

11.6 Environmental Discussion

The lockdown comparison provides an opportunity to understand the relationship between human activities and environmental pollution.

Reductions in transportation and industrial operations may have contributed to lower concentrations of primary pollutants.

However, air quality is also influenced by natural and meteorological factors.

Consequently, the study considers statistical evidence alongside environmental conditions rather than attributing every observed change exclusively to lockdown restrictions.

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12. CONCLUSION

This case study presents an exploratory and statistical analysis of urban air quality in Indian cities between 2015 and 2020.

The study focuses on PM2.5 and NO_2 concentrations to understand pollution distributions, city-wise variations and seasonal patterns.

Pandas is used for data processing and descriptive analysis, while SciPy and Statsmodels support confidence interval estimation and hypothesis testing. Seaborn provides graphical representations of pollution distributions and relationships between pollutants.

The comparison between the corresponding periods in 2019 and 2020 helps investigate changes in air pollution during the COVID-19 lockdown.

Paired t-tests, independent t-tests or Mann–Whitney U tests are selected according to the structure and assumptions of the data.

Correlation analysis further provides insights into relationships between different pollutants. The study also highlights the importance of distinguishing statistical association from causation.

Overall, this project demonstrates how statistical data analysis can support environmental research, urban air quality monitoring and evidence-based pollution-control policies.

The study contributes to the objectives of SDG 11 and SDG 13 by promoting healthier urban environments and encouraging data-driven environmental management.

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